# Brightlearn Research Assignment 1

## Section A: Data Fundamentals

1. A database is a place where data is stored in an organized way so it can be easily accessed and managed.

/ Databases are important because almost everything today uses data like apps, banks, and schools without data it will be hard to keep track of information.

/ A Bank that stores customer accounts & transactions
/ Schools store student records & marks.

2. Data Warehouse is a central, big database that stores different databases that have large amounts of historical data for analysis.

/ The difference is that the Database is used for daily tasks & a Data Warehouse is used for analyzing data.

/ A Company may use a data Warehouse to make better decisions & identify trends.

3. A data lake is a storage that stores raw data in its original form.

/ The difference is that a Data Warehouse stores clean, structured data and a Data lake stores raw, messy data.

/ A business would prefer a data lake over a data Warehouse when it wants to keep all data for future use.

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4., A Data Mart is a Smaller part of a data Warehouse focused on a Specific department

, it is usually used by teams like Sales / marketing.

5., Structured data: is a data that is organized and in a tabular Format [Can be stored in a table]
Example: Spreedsheets (excel)

/ Semi-Structured data is data that is partially structured but not fully structured, it has its own structure not the tabular format.
Example: JSON file.

/ Unstructured data is data that is unorganized and cannot be stored in a tabular format [it is hard to capture].
Example: music / Videos or images

6., Data Modeling is a process of designing how data will be stored and organized.

/ It is important because it makes systems efficient and easy to use.

/ The two types of data models are Conceptual model and Logical model. (bigger picture & detailed structure).

7., Data Mining is finding patterns in data.

/ In a business it is used to predict customer behavior & improve marketing.

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The two examples are Online Stores recommending products & Banks detecting fraud.

8. A DBMS is a software that is used to manage & manipulate data.

Examples:
MySQL → this is for websites
PostgreSQL → for advanced data systems
Oracle → for large companies
MongoDB → this is for flexible data (NoSQL)

9. Extract → gets the data
Transform → clean / organize the data
Load → Stores the data

It is important because it prepares data for analysis.

10. Data storage formats:
CSV → for simple tables
JSON → for APIs & web Data
Parquet → for big data
Avro → for data streaming

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## Section B : AI & Machine learning

11., AI it is when machines are being designed to think and act like humans

, Narrow AI does one task ( Voice Assistants)
General AI can do anything humans can (it is not fully developed yet)

12., Machine Learning it is when computers learn from data instead of being programmed step-by-step.

, Supervised ML -> learns from labeled data
, Unsupervised ML -> ~~tea~~ finds ~~pop~~ patterns
, Reinforcement -> learns through reward.

13., Deep learning is a type of ML that uses neutral networks.

, Neutral networks work like the human brain - they process information in layers.

14., NLP helps the computer understand human language.

, Examples: Voice assistance, Chatbots, Translators (apps)

15., LLM is a Large language Model that understands & generates text.

, Examples: ChatGPT (Conversations), Claude (for Writing & analysis)

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16., Generative AI creates new content like text, images, or music.

, Examples: Chatgpt [for text], DALL-E [for images], Midjourney [for art].

17, Training Dataset is the data that is used to train an AI model.

, If the data is bad/biased, the AI will give wrong or unfair results.

18., Computer Vision allows machines to understand images

, Examples are face recognition & Self-driving cars.

## Section C: APIs & Tools

19., API is a way for different systems to communicate with each other.

, Example: In a restaurant you tell ~~a wait~~ a waiter what you want, they take your request to the kitchen and bring your food back. You don't need to know how the kitchen works, the waiter handles everything. The API takes a request from one system and returns a response from another just like waiters.

, They are important because they allow apps and systems to connect & share data easily.

20., API Key is like a password that gives access to an API. It helps to identify who is using the API & controls usage.

, It should be stored safely. If it is exposed, someone else could use it, which might lead to data theft, extra costs, or security problems.

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21. REST stands for Representational State Transfer. It is a common way of building APIs.

/ Main methods:
GET → Used to fetch data.
POST → Used to create new data.
PUT → Used to update existing data
DELETE → Used to remove data.

22. JSON is a simple format used to store & exchange data.

/ it is popular because it is easy to read and works well with APIs.

/ Example:
{ "name": "Samkelo",
"age": 21,
"course": "Data ~~Sci~~ Science"
}.

23. Prompt Engineering is a skill of asking AI questions clearly to get better answers.

/ It is important because better prompts lead to better results.

/ Example 1:
Bad: "Tell me about AI"
Good: "Explain AI in simple terms & give 2 real-life examples"

Example 2:
Bad: "Write code"
Good: "Write a Python program that calculates the average list of numbers"

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24, A map function applies the same operation to every item in a list.

, Example (Python):

```
numbers = [1,2,3]
result = list(map(lambda x: x * 2, numbers))
print(result)   # Output: [2,4,6]
```

25, Cloud Computing means storing & accessing data over the internet instead of your computer.

, Types:

SaaS → ready-to-use software (Google Docs)
IaaS → provides infrastructure (AWS)
PaaS → provides platform to build apps (e.g., Heroku)

26, ~~Cloud~~ Version Control is a system that tracks changes in a ~~cog~~ code.

, Git is widely used because it helps developers work together & manage change easily.

, Commit saves changes & Branch is for creating a separate version to work on.

27, Jupyter Notebook is a tool used to write & run code while also showing results, Graphs & explanations.

, It is mostly used with python and is popular because it makes data analysis easier to understand and present.

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28. / Python is a programming language that is easy to learn & is very powerful for data and AI.

/ Libraries:
Pandas → Used for data analysis
NumPy → Used for calculations
Matplotlib → Used for graphs
Scikit-learn → used for machine learning.

## Section D: Data Ethics, Governance & Careers

29. / Data Privacy is about protecting personal information.

/ It is important because people's data should not be misused.

/ GDPR (General Data Protection Regulation) is a law in the European Union that protects people's data and gives them control over how it is used.

30. / Data Ethics means using data responsibly.

/ Examples:
Selling user data without permision → Violates privacy.
Biased AI decision → unfair treatment.
Spreading fake information → misleads people.

31. / Data Governance is how an organisation manages & protects it's data.

/ Key parts are data Security, data quality, rules & policies.

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32. Bias in AI happens when AI system gives unfair results based on the data it was trained on.

Example: A hiring system that prefers certain genders, this can lead to discrimination & unfair decisions.

33.:

| Role | What they do | Tools |
|---|---|---|
| Data Analyst | Analyze data | Excel, SQL |
| Data Scientist | Build models | Python, ML |
| Data Engineer | Manage data systems | SQL, Spark |
| AI Engineer | Build AI systems | Python, TensorFlow |

34. BI is a process of using data to make better business decisions.

Tools:
Power BI (Creates dashboards)
Tableau (Visualizes data clearly)

## Section E: Personal Reflection

35. I found AI the most interesting because it shows how machines can learn and think in a way similar to humans.

36. Data Modeling was the most difficult for me because it involves understanding how data is structured. I used YouTube videos & examples to understand it better

37. Data and AI will affect my future by making tasks faster & more efficient. For example AI tools can help to analyze data quickly, and chatbots can improve customer service.

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